Kinetic energy and net work

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect net work to a change in kinetic energy.

SUCCESS CRITERIA - I can...

1. Calculate kinetic energy with speed squared.
2. Use net work rather than one force's work.
3. Connect the sign of work to a speed change.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use "kinetic energy" and "work" to explain your model or result. Sentence starters are optional.

kinetic energy: energy associated with motion

work: energy transferred by a force along a displacement

initial: at the beginning of the interval; subscript i

01 Warm-up: bring back an earlier idea

At fixed momentum magnitude, mass becomes 3 times as large. Derive the kinetic-energy multiplier using $K=p^2/(2m)$.

Word help: momentum = mass times velocity, including direction

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $K = \frac{1}{2}mv^2$; $W_{net} = K_f - K_i$.
i = initial; f = final. Positive net work increases kinetic energy.

Worked reasoning: A 2 kg cart at 2 m/s has K=4 J. Adding 5 J of net work gives final K=9 J, so final speed is 3 m/s.

Your turn: Draw before-and-after energy bars and label the transfer between them.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

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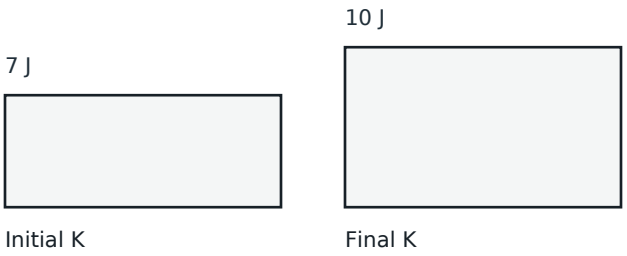
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

A lab cart’s kinetic energy changes from 7 J to 10 J, as shown.
Find the net work.
Explain how its speed changes if mass stays constant.

Sentence starter: Under the assumption …, the evidence … supports … because … .



Marzano check - circle: 1 2 3

04 Apply the physics

A 4 kg cart moves at 2 m/s. The net work on it is +24 J. Complete the energy account, then find its final kinetic energy and speed.

Sentence starter: I found … [unit]. This value means … in this situation.

Initial	Transfers	Final

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

05 Explain your reasoning

A student adds the work of a pull but ignores friction. Why might their predicted speed be too high?

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Two equal 4 kg carts collide elastically along one line. A starts at +6 m/s; B starts at rest.
Find both final velocities under the ideal model.
Use before and after snapshots to organize the momentum account.

Word help: ideal = using stated simplifying conditions

Sentence starter: I found … [unit]. This value means … in this situation.

Before	After

Label mass, velocity and momentum.

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

A 2 kg object slows from 6 to 2 m/s.
Find net work.
Use the energy model to show the quantities you compare; label units and any unknowns.

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 3 kg object starts at 4 m/s and receives -18 J of net work.
Find final speed. What is the most negative net work possible before it first stops?
Use the energy model to show the quantities you compare; label units and any unknowns.

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3